# AURANT ACEX: A Multimodal Quantitative Trading Architecture Integrating Temporal Fusion Transformers and Mixture-of-Experts (Part 2: Methodology, Procedures, and Data)

Quantitative Research Division

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## 1 Methodology

The AURANT ACEX architecture is built upon a mathematically rigorous foundation, combining representation learning, dynamic routing, and uncertainty quantification.

### 1.1 Temporal Fusion Transformer Backbone

The core sequence modeling is handled by a modified Temporal Fusion Transformer (TFT). Given a multivariate time-series input $\mathbf{X}_t \in R^{d_{input}}$, the system first maps the inputs into a latent space via a Variable Selection Network (VSN). The VSN computes a set of dynamic weights:

$$v_t = \text{Softmax}\left(\mathbf{W}_{vsn}\mathbf{X}_t + \mathbf{b}_{vsn}\right) \tag{1}$$

The weighted features are then processed by a Gated Residual Network (GRN), which allows the model to skip irrelevant transformations.

The temporal dependencies are captured using Rotary Position Embeddings (RoPE) within a multi-head self-attention module. For queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$, the attention is computed as:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{2}$$

where $d_k$ is the dimensionality of the key vectors. To prevent lookahead bias, a strict causal mask is applied, setting the upper triangle of the attention score matrix to $-\infty$.

### 1.2 Time-Series Mixture-of-Experts (MoE)

To handle non-stationary regime shifts, AURANT ACEX implements a MoE layer consisting of $E = 8$ distinct expert networks. Let $E_i(\mathbf{h}_t)$ represent the output of the $i$-th expert for hidden state $\mathbf{h}_t$. A gating network computes routing probabilities:

$$g_t = \text{Softmax}\left(\mathbf{W}_g\mathbf{h}_t\right) \tag{3}$$

To ensure computational efficiency and avoid representation collapse, the model utilizes Top-$k$ routing (where $k = 3$):

$$\mathbf{y}_t = \sum_{i\in\text{Top-}k(g_t)} \frac{g_{t,i}}{\sum_j g_{t,j}} E_i(\mathbf{h}_t) \tag{4}$$

A load-balancing auxiliary loss $\mathscr{L}_{load}$ is introduced to penalize the routing network if it disproportionately favors a single expert.

### 1.3 Loss Formulation and Optimization

The model is trained using a composite loss function to balance point forecasting accuracy, distributional robustness, and rank correlation.

$$\mathscr{L}_{total} = \lambda_1\mathscr{L}_{huber} + \lambda_2\mathscr{L}_q + \lambda_3\mathscr{L}_{bce} + \lambda_4\mathscr{L}_{ic} \tag{5}$$

Where $\mathscr{L}_{huber}$ is the asymmetric Huber loss targeting the conditional mean, $\mathscr{L}_q$ is the pinball loss for quantile estimation, $\mathscr{L}_{bce}$ focuses on directional accuracy (sign prediction), and $\mathscr{L}_{ic}$ explicitly maximizes the soft Information Coefficient.

## 2 Procedures and Data

### 2.1 Data Processing Pipeline

The system ingests structured market data stored in compressed Parquet format, ensuring scalable, memory-safe chunked I/O. The raw data encompasses OHLCV (Open, High, Low, Close, Volume) metrics, macroeconomic signals, and cross-sectional rankings.

Feature engineering is executed with strict causal boundaries. Features include multi-scale entropy to detect regime complexity, Volatility-Adjusted Momentum, Amihud/Kyle illiquidity proxies, and fractional differentiation. For fractional differentiation, the series is transformed using the binomial series expansion:

$$(1-B)^d x_t = \sum_{k=0}^{\infty} \binom{d}{k} (-1)^k x_{t-k} \tag{6}$$

where $B$ is the backshift operator and $d$ is a non-integer differencing order, preserving memory while achieving stationarity.

## 2.2 Walk-Forward Backtesting

To validate the model, we employ a rigorous walk-forward validation framework. The time-series is partitioned chronologically into expanding training windows and fixed-size out-of-sample (OOS) test folds. Crucially, an explicit "purge gap" $\Delta = \max(h)$ (where $h$ is the prediction horizon) is enforced between the end of the training set and the start of the validation/test set. This mathematically guarantees the elimination of data leakage and lookahead bias.